Data acquisition and preprocessing

Vegetation
MODIS EVI and LAI
Greenness and canopy structure

Climate
VPD, LST and soil moisture
Precipitation and temperature
ERA5-Land and MODIS

Surface water
Occurrence / seasonality
JRC Global Surface Water Version 1.4

Landscape
Elevation, slope, land cover
WorldClim temperature and precipitation

Temporal alignment and valid-record filtering 0.1° grid-year panel China, 2018–2023

Indicator construction and aquatic classification

Vegetation anomalies
Local seasonal standardization
z = (value − local mean) / local SD
Late-summer EVI and LAI
Event: z < −0.5
Combined event: EVI OR LAI

Compound heat–dryness
Summer standardized components
HDSI = (VPDz + LSTz − SMz − Precipz) / 4
Low exposure: HDSI < Q25
High exposure: HDSI > Q75

Aquatic contexts
Permanent: occurrence ≥80% and seasonality ≥10 months
Seasonal: 1–<10 months
30 m UTM distance, 1 km centroid rule
Permanent / seasonal / other

Vegetation risk and aquatic fragility buffering

Exposure groups
Low and high HDSI within each aquatic context

Event probability
Risk = event count / group sample size

Vegetation fragility
F = Risk(high HDSI) − Risk(low HDSI)

Aquatic contrast
AFB = F(comparison) − F(permanent)

Diagnostic analyses and supplementary checks

Exposure and canopy
Stress components under high HDSI
Late-summer minus spring LAI z
Low-to-high event probabilities

Regional comparisons
Macroregion stratification
AFB by region and response
Context and exposure support

Supplementary checks
Environmental adjustment
Landscape strata and crop exclusion
Stress, threshold and year sensitivity

Uncertainty estimation: paired grid bootstrap Grid-cluster model intervals 1°, 2° and 5° spatial blocks